Crispat parallelization — USL fits and empirical phase attribution

Schraivogel dataset, batch1 (172 gRNAs total: 86 gRNAs × 2 methods), omnibenchmark node (128 logical cores), `OMP_NUM_THREADS=1`, `--no-plots` enabled, instrumented branch `parallel-gauss-instr`.

USL fits ($S(n) = n / (1 + \alpha(n-1) + \beta \cdot n(n-1))$)

Method	α (serial)	β (coherence)	n*	S* (USL)	Observed peak	Δα vs E1	ΔS* vs E1
`ga_gauss`	0.68 %	2.20×10^{-4}	66.6	27.5 ×	26.31 × @ n=100	−43 %	+24 %
`ga_poisson_gauss`	1.36 %	3.20×10^{-4}	55.3	20.4 ×	19.87 × @ n=64	+37 %	−6 %
run (combined)	0.93 %	2.60×10^{-4}	61.8	24.3 ×	23.51 × @ n=64	—	—

E1 baseline (with per-gRNA matplotlib plots): ga_gauss α=1.20 %, β=2.80×10^{-4}, S=22.2×.*

Empirical phase attribution at n = 64

Phase	Wall (s)	Share of run	USL role
`load_h5ad`	0.04	0.3 %	α (true serial)
`pool_setup`	1.74	11.8 %	β — linear in n_jobs
`fit_loop`	11.86	80.6 %	parallelisable
`assemble`	0.32	2.2 %	α (true serial)
`write_csv`	0.09	0.6 %	α (true serial)
`combine_batches`	0.06	0.4 %	α (true serial)
Total non-fit (serial floor)	**2.25**	**15.3 %**	α + β
Run total	**14.70**	100 %	

The USL fit predicted α = 0.93 % for the combined run. Sum of non-fit_loop phases at n=64 is 2.25 s / T_1 (345.5 s) = 0.65 %; at n=100 it is 3.34 s / 345.5 s = 0.97 %. The model's α is the empirical serial-floor fraction.

`pool_setup` is β

n_jobs	2	4	8	16	32	64	100
`pool_setup` (s)	0.07	0.14	0.34	0.50	0.88	1.74	2.84
per-worker (ms)	35	35	43	31	27	27	28
share of run total	0.05%	0.20%	0.76%	1.67%	5.63%	11.84%	18.61%

Linear in n_jobs at 28 ms per worker spawn (fork + COW dirtying + pyro/torch import). At n=100 this single phase is 19 % of total wall. Concretely the β term made visible.

Memory & contention checks (denet eBPF)

n_jobs	Peak RSS (GB)	Per-worker (MB)	Off-CPU total (s)	Major page faults
1	0.6	—	125.8	4
8	4.1	437	147.4	0
32	14.9	445	48.2	8
64	29.1	445	41.9	26
100	43.7	431	40.0	91

Findings:

- *Per-worker RSS flat at 440 MB (no leak; pure fork-COW cost).*
- *Off-CPU time **decreases** with n_jobs — workers are CPU-bound at scale, not blocked. Rules out lock contention.*
- *Major page faults stay under 100 over the whole run on a 256 GB node. No memory pressure.*
- *Minor page faults plateau at 6 M for n ≥ 32: fork-COW fault budget fully amortised.*

Unexplained: per-gRNA `fit_one` wall grows with n_jobs

n_jobs	2	4	8	16	32	64	100
median fit_one (s)	1.73	1.70	1.82	2.41	2.46	3.62	4.70
p95 fit_one (s)	1.97	2.04	2.88	5.36	3.63	5.30	6.09

Per-gRNA work is identical, yet wall doubles+. Off-CPU rules out blocking; page-fault plateau rules out fault traffic; CPU saturation is near n_jobs × 100 %. Strongest remaining hypothesis: NUMA cross-socket traffic (workers reading parent-resident pages from the other socket). Test queued as E1c — pin all workers to one socket via `os.sched_setaffinity` in `_init_worker` and re-measure.